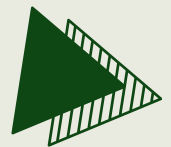


Citi Bike Usage Distribution on Manhattan

Finding the most dense biking place & more



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Research Question & Objective

This project is initiated to understand the general usage pattern of Citibikes in Manhattan.

With this premise, we want to find out ***which neighborhood people use Citibikes the most and what could be the factor behind that.***

The objectives of this project will thus include:

- Measure bike usage density by neighborhood
- Analyze trip duration patterns within neighborhoods
- Assess differences between electric and classic bike usage
- Provide insights to inform future policy and infrastructure planning

Data Description

This project uses the NYC Citi Bike Usage Data dataset and focuses on the data from January 2024. Though there might be fewer people riding bikes during winter season, the result can certainly inform the necessity of bike usage in a neighborhood. The key variables involved in this analysis are: (1) Trip duration calculated through start and end time; (2) Bike type (electric vs. classic); (3) Start station locations and numbers.

Methodology used in this project includes:

- Data preprocessing and cleaning
- Data aggregation spatially and temporally
- General bike usage pattern comparison across neighborhoods

We assume, the usage would be highest in central business areas, with electric bikes showing shorter trip duration on average

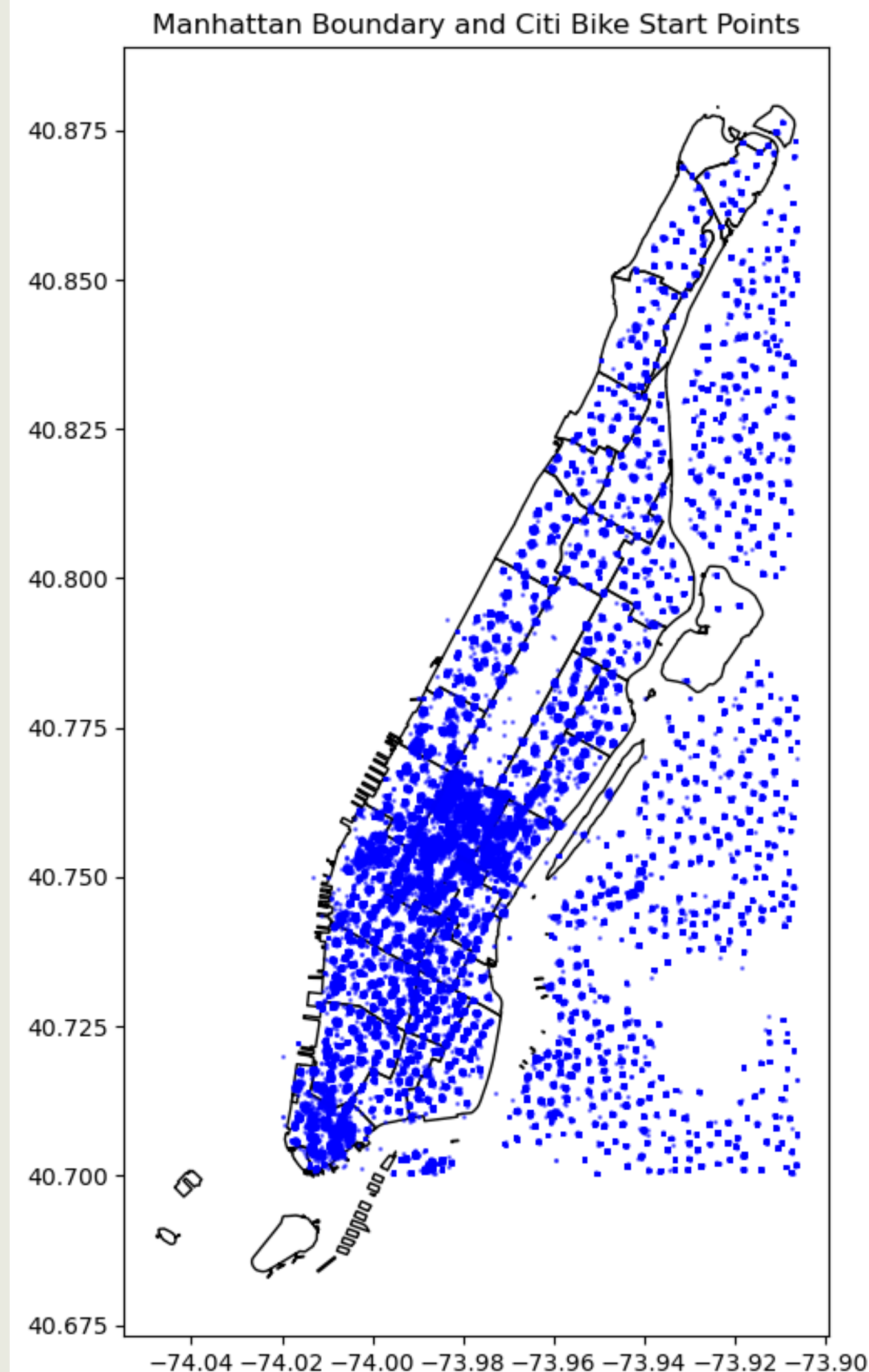
Usage Duration & Distance

	Avergae riding duration (min)	Average riding distance (km)
Classic bike	11.5	1.4
Electric bike	10.2	1.8
Summary	The average riding duration of classic bike is longer than electirc bike (though not significantly longer)	The average riding distance of electric bike is longer tna classic bike

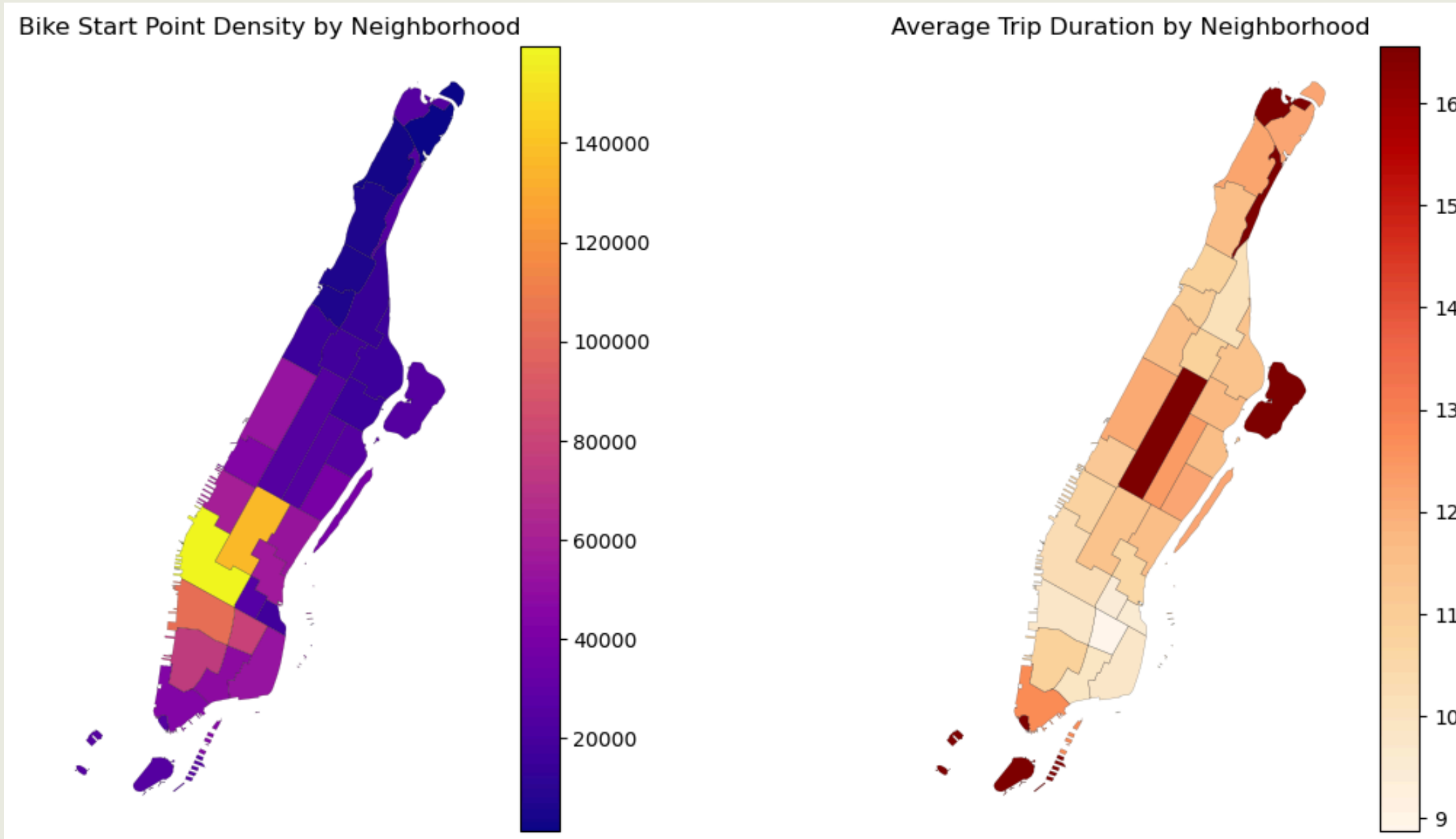
**As electirc bike might be easier and faster to ride than classic bike, and could be especially easier to use on hilly Manhattan, it seems like riding electirc bike can be more efficient than classic bike, but without more detail analysis, it's hard to say if people would delibratly pick electric or classic bike, as the differences is not significant.*

Usage density on map

By showing the start point of each citi-bike user data, we can see that downtown, especially Financial District and Midtown has more bike usage clusters thus higher usage density. Meanwhile, the usage density decreases significantly from mid to uptown areas. This pattern may match with the income level (where lower Manhattan has higher median household income), more dense in population. People living in downtown area are more willing to ride to work or surrounding due to the convenience, while uptown residents might prefer taking Metro as the travel distance could be longer for them.



Bike Density & Duration



Bike Density by Neighborhood

The choropleth map for bike density further indicate the significant cluster of bike usage density in neighborhood include Midtown and Chealsea, where while walkable, biking could be a effective way to navigate the area. We can also see the Inwood area has significantly low density, as the area could be more hilly, and more suburban, where driving may be more convinient.

Trip Duration by Neighborhood

But its very interesting that when come to the trip duration, the most densed bike usage area become less significant. The most read areas are almost all park or tourist areas such as the Central Park, Randals Island, Battery Park... This also make sense, because these are all area that would encourage visitors to have biking experience, and people spend more time navigating. Furthermore, the bike usage density seems to be inversely proportional to the trip duration of the area.

Temporal Variations in a day

Peak Usage

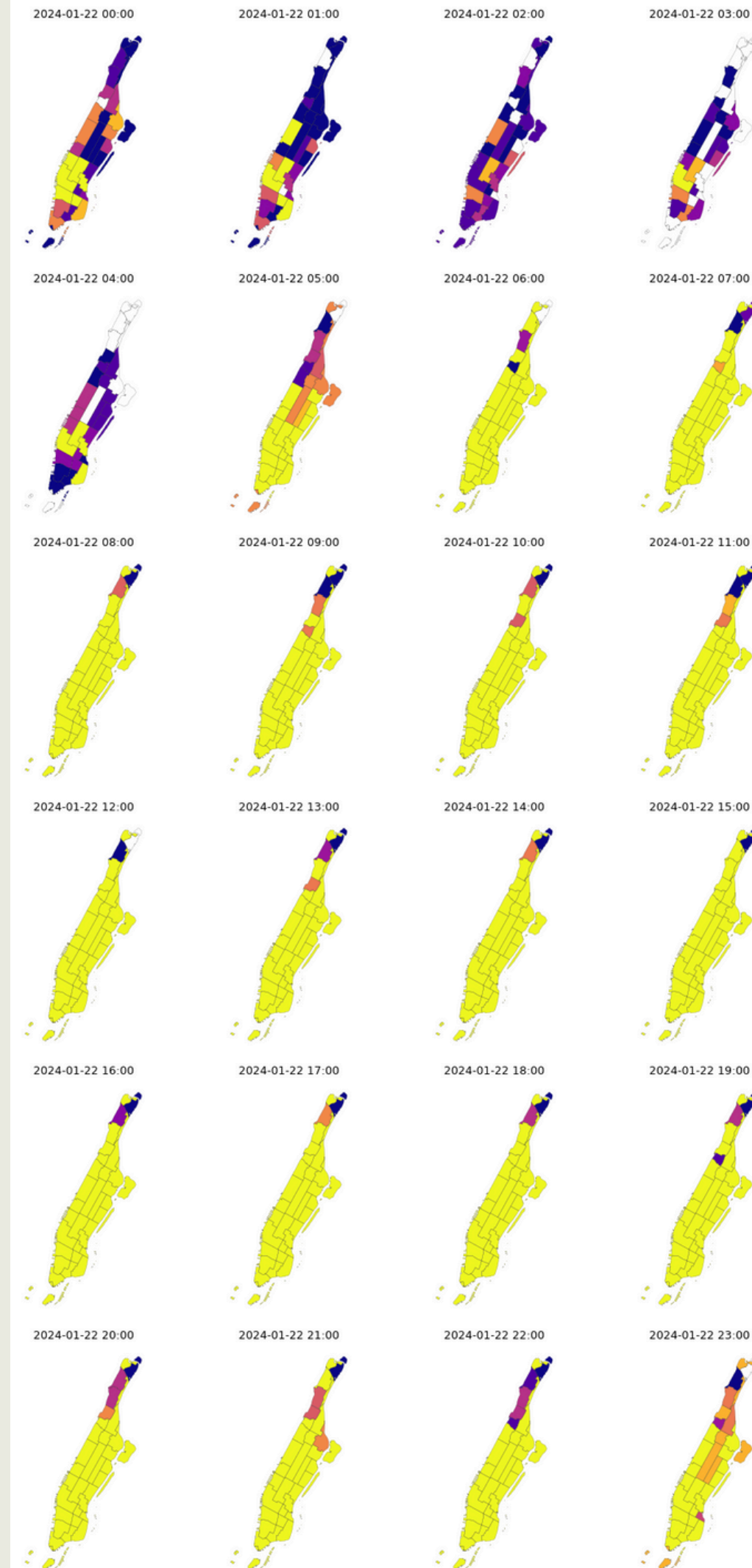
A morning peak of bike usage from 6AM-9AM, High-density areas during these hours are concentrated in Midtown and Downtown Manhattan, reflecting commuting patterns. This suggests that many people are using Citi Bikes to travel to work or as transit hubs in these business-heavy neighborhoods.

A second peak in bike usage density begins in the late afternoon around 4 PM, continuing into the early evening, peaking again around 6-7 PM.

Here is a temporal variations diagram showing how usage density (purple to yellow = low to high) change during certain time of the day. This 24h variation on Jan.22, a regular Monday, can show you how time and location may be related to the general bike usage pattern in the neighborhood, and the city in general.

Possible Factors

- ***Business and Commercial Hubs:*** The highest density consistently appears in business districts, particularly Midtown and Lower Manhattan.
- ***Commuting Patterns:*** Morning and evening peaks suggest that Citi Bikes are popular with commuters, likely as an alternative or supplement to public transit.
- ***Recreational and Late-Night Usage:*** Moderate density in neighborhoods outside of business hubs, particularly during off-peak hours, points to possible recreational usage. Areas with entertainment venues, parks, or nightlife attract users outside traditional commute hours.



Oppurtunities for future development

Enhanced Bike Availability in High-Density Areas	Dynamic Resource Allocation	Recreational Area Focus	Infrastructure Improvements
(1) Prioritize Citi Bike availability in Midtown and Downtown during peak hours to meet commuter demand. (2) Consider increasing the number of electric bike docks in these areas to support faster, shorter trips preferred by commuters.	(1) Implement a data-driven, real-time allocation system to relocate bikes from lower-demand areas to high-demand locations during peak hours, optimizing system efficiency.	(1) Ensure bike availability in parks and tourist spots where usage duration is longer, accommodating recreational and tourist demand.	(1) Explore expanding bike lanes or dedicated bike paths in business and entertainment districts, supporting safe and efficient bike transit in areas with high bike density.

To align with commuter demands, the Citi Bike program should consider expanding both the number of bikes and dock availability in Midtown and Downtown Manhattan, particularly targeting early morning and late afternoon peak hours. Implementing a system that monitors bike availability in real-time could allow for dynamic reallocation, moving bikes to these high-demand areas as needed. Furthermore, given the patterns we observed in recreational zones—where riders tend to take longer trips—it would be beneficial to invest in more durable bike docks and support infrastructure in these areas. This approach would cater to both commuters and leisure riders, balancing accessibility across neighborhoods with varied usage patterns.

Next Steps

To align with commuter demands, the Citi Bike program should consider expanding both the number of bikes and dock availability in Midtown and Downtown Manhattan, particularly targeting early morning and late afternoon peak hours. Implementing a system that monitors bike availability in real-time could allow for dynamic reallocation, moving bikes to these high-demand areas as needed. Furthermore, given the patterns we observed in recreational zones—where riders tend to take longer trips—it would be beneficial to invest in more durable bike docks and support infrastructure in these areas. This approach would cater to both commuters and leisure riders, balancing accessibility across neighborhoods with varied usage patterns.

To refine resource planning, future analyses should incorporate seasonal data to account for shifts in usage patterns throughout the year. Additionally, considering user demographic data could offer valuable insights to tailor infrastructure investments more effectively, aligning with specific community needs and enhancing overall user experience.

Conclusion

Our analysis highlights that Citi Bike usage peaks in Midtown and Downtown Manhattan, driven largely by commuter traffic during the busiest hours of the day—morning and evening. These central areas show the highest density of bike starts, reflecting their role as business and transit hubs where people rely on bikes for quick connections. Outside of peak commuting times, bike usage shifts slightly, with a noticeable dip midday and increased usage again in the late afternoon. Recreational areas, including parks and tourist hotspots, stand out with longer trip durations, indicating that these locations serve more leisurely riders or tourists rather than the quick, functional trips typical of business districts. This distribution pattern underscores the need to prioritize bike availability in high-density areas during peak hours while also ensuring sufficient resources in recreational zones to accommodate longer rides and casual use.

Interpretive exercise for the city to optimize citi bike resource allocation

Considerations should include

- **Commuter Focus:** Where would you prioritize placing bike docks within this new business district to best support commuting patterns? How would you ensure bike availability during these peak times?
- **Electric vs. Classic Bike Allocation:** What proportion of electric to classic bikes would you recommend? Would you adjust this mix for recreational zones within the district?
- **Recreational and Off-Peak Access:** How would you adapt the bike allocation and dock placements to accommodate leisure riders who may use bikes for extended periods?
- **Dynamic Resource Reallocation:** How might you implement a dynamic reallocation strategy to adjust bike availability throughout the day? What operational or logistical considerations would you need to keep in mind?
- **Infrastructure Considerations:** What infrastructure improvements (e.g., bike lanes, signage) would you recommend to enhance safety and accessibility for both commuters and recreational users in this new district?

Exercise Agenda

(1) Group Discussion: Participants shall be divided into small groups, each responsible for developing a resource allocation strategy that addresses the questions above.

(2) Present the Findings: Each group will present their strategy, including their reasoning behind dock placements, bike type allocation, and any infrastructure recommendations.

(3) Discussion: Facilitate a full-group discussion comparing strategies, focusing on different approaches to balancing commuter and recreational needs, as well as logistical challenges.

Through this exercise, we would like the participants to walk away with a practical understanding of how to apply Citi Bike usage insights to resource allocation decisions in new development areas. This exercise will help solidify their grasp of the data, encourage creative problem-solving, and foster a strategic approach to planning for sustainable urban mobility.